Chapter 35 — Local Full-Text Search (FTS)

> **Part IV — Offline-First & Synchronization** · Complexity: **Standard** (≥25 interview questions)
>
> **Prerequisites:** Ch 3 (Room/SSOT), Ch 9 (Debounced Search), Ch 10 (Suggestions/ autocomplete), Ch 29 (offline-first). Enables the "offline search" and "local prefix match" deferred from Ch 9/10.
>
> **Reference surface in StreamCart:** searching the cached catalog, watch history, and saved items **offline**, instantly — and merging local results with remote search (Ch 9/10).

35.1 Problem Definition and Real-World Context

Ch 9/10 built **remote** search (debounced network queries). But an offline-first app (Ch 29) must also search its **local** data — and do it well. Naive local search fails:

- **`LIKE '%query%'` doesn't scale or rank.** A substring `LIKE` scan over a large table is slow (no index use for leading wildcards), can't rank by relevance, and handles neither word boundaries, stemming, nor multi-term queries.
- **No offline search.** With only remote search, the app can't search cached data on a train — breaking offline-first.
- **Poor relevance.** Users expect matches ranked by relevance (title match > description match), prefix/typeahead matching, and multi-word queries — `LIKE` gives none of that.
- **No highlighting.** Search UIs highlight the matched terms; `LIKE` can't produce snippets.

SQLite FTS (Full-Text Search), exposed via Room's `@Fts4` / `@Fts5`, builds an **inverted index** over text columns, enabling fast tokenized search with **relevance ranking** (BM25), **prefix** and **phrase** queries, and **snippet/ highlight** support — all **local and offline**. The architecture is about **what to index**, **FTS4 vs FTS5 + external-content tables**, **query syntax/ranking**, **keeping the index in sync** with the source table, and **merging local + remote** results (Ch 10).

Three practical use cases

1. **E-commerce (StreamCart).** Instant offline search over the cached product catalog (title/brand/description), ranked, with prefix matching for typeahead (merged with remote, Ch 10).
2. **OTT (Watch).** Search watch history and "my list" offline by title/cast.
3. **Productivity/Social (Social).** Search local messages/notes/drafts offline with highlighting.

Why this is architectural

Local FTS defines **offline search capability**, the **index design** (what's searchable, tokenizer, external vs contentless), **relevance/ranking**, and how local results **compose with remote** — properties that determine whether search works offline and feels good.

35.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Full-text search cached data locally/offline, tokenized (not substring).
F2	Rank results by relevance (BM25); weight fields (title > description).
F3	Support prefix (typeahead) and multi-term/phrase queries.
F4	Produce highlighted snippets of matches.
F5	Keep the FTS index in sync with the source (Room) table.
F6	Merge local results with remote search (Ch 10).

Non-functional requirements

Attribute	Policy
Performance	Sub-frame local query over large tables; indexed, not scanned.
Storage	Index bounded; external-content to avoid duplicating data.
Freshness	Index updated on data change (triggers / offline sync).
Relevance	Ranked; field-weighted; tokenizer handles case/diacritics.
Offline	Works with no network (Ch 29).

Scope

- **In scope:** Room FTS (`@Fts4` / `@Fts5`), external-content tables, tokenizers, MATCH queries (prefix/phrase/boolean), BM25 ranking, snippet/highlight, index sync, merging local+remote.
- **Out of scope:** remote search/debounce (Ch 9/10), server-side search/relevance (backend), offline sync (Ch 29–33).

Assumptions & constraints

- Room 2.x with FTS support; SQLite FTS4/FTS5 (FTS5 preferred where available). Source data cached in Room (Ch 3). `minSdk 24` (FTS5 generally available on modern Android SQLite). Prefer **FTS4 with Room** for broad compatibility, FTS5 where the platform SQLite supports it.

35.3 System Design Fundamentals

- **Inverted index.** FTS builds a term → documents index, so a query matches by **tokens** (words), not substrings — fast and relevance-rankable, unlike `LIKE`.
- **Room FTS entities.** Annotate a table with `@Fts4` (broadly supported with Room) or `@Fts5`; Room creates the virtual FTS table. Query with `MATCH`.

- **External-content tables.** To avoid **duplicating** the data (the FTS table storing its own copy), use an **external-content** FTS table that indexes an existing Room table (the `content=` option) — the index references the source rows, saving storage; you keep the index in sync via triggers or Room DAO writes.
- **Tokenizers.** Control how text is split/normalized: `unicode61` (default, Unicode-aware, case-folding), with options like `remove_diacritics` for accent-insensitive search. The tokenizer determines matching behavior (case, diacritics, word boundaries).
- **Query syntax (MATCH).** Supports **prefix** (`shoe*`), **phrase** (`"running shoe"`), **boolean** (`shoe AND red` , `NOT`), and **column filters** (`title: shoe`). Prefix enables typeahead; sanitize user input to valid FTS syntax.
- **Ranking (BM25).** FTS5's `bm25()` ranks by relevance; you can **weight columns** (title weighted higher than description) to tune relevance. FTS4 offers `matchinfo` / `offsets` for custom ranking.
- **Snippet / highlight.** `snippet()` / `highlight()` (FTS5) produce excerpts with the matched terms marked — for search-result highlighting.
- **Index sync.** The FTS index must reflect the source table: update it when rows change (Room triggers, or write both in the repository; external-content tables need explicit sync on insert/update/delete).
- **Local + remote merge (Ch 10).** Offline-first search shows **local FTS results instantly**, then merges **remote** results when online — the same local-first composition as Ch 10 suggestions.

Core insight: **index searchable text in a Room FTS table (external-content to save storage), query with MATCH for tokenized prefix/phrase search ranked by BM25 with field weights, keep the index in sync, and merge local-first with remote** — giving fast, ranked, offline search that `LIKE` can't.

35.4 High-Level Architecture



Components & ownership

- **Source table** (Room SSOT) holds the data (cached, Ch 3/29).
- **FTS table** (external-content) is the inverted index over its text columns.
- **Index sync** keeps FTS in step with the source (triggers / repo).
- **Query** sanitizes input to MATCH syntax, runs ranked search with snippets.
- **Merge** composes local (instant, offline) + remote (Ch 10).

Direction: data → FTS index; query → MATCH (ranked) → results → merge with remote → UI.

35.5 Detailed Component Design



Component	Responsibility	Note
source entity	the data	Room SSOT (Ch 3)
FTS entity (`@Fts4` / `@Fts5`)	inverted index over text cols	external-content (no data dup)
DAO `MATCH` query	ranked search + snippet	BM25 + field weights
index sync	keep FTS current	triggers / repo writes
sanitizer	user text → valid FTS syntax	prefix, escape specials
merge (Ch 10)	local-first + remote	offline-first search

35.6 End-to-End Data Flow

Offline local search



Index sync on data change



35.7 Complete Kotlin Implementation

Package: `com.streamcart.data.catalog.search`.

35.7.1 Source entity + FTS entity (external-content)

```
@Entity(tableName = "products")
data class ProductEntity(
    @PrimaryKey val id: String,
    val title: String,
    val brand: String,
    val description: String,
    val priceCents: Long,
)

// External-content FTS4 table: indexes ProductEntity's text columns WITHOUT duplicating data.
@Fts4(contentEntity = ProductEntity::class)      // Room wires content= to the source table
@Entity(tableName = "products_fts")
data class ProductFts(
    val title: String,
    val brand: String,
    val description: String,
    // rowid links to ProductEntity's rowid (Room manages this for content tables)
)
```

35.7.2 Search DAO (MATCH, ranking, snippet)

```
data class ProductResult(val id: String, val title: String, val snippet: String)

@Dao
interface ProductSearchDao {
    // Join FTS match back to the source table; rank by relevance.
    // (FTS4 uses matchinfo/offsets; this shows the general shape. FTS5 adds bm25() +
snippet().)
    @Query("""
        SELECT p.id AS id, p.title AS title,
               snippet(products_fts, '[', ']', '...', -1, 12) AS snippet
        FROM products_fts
        JOIN products AS p ON p.rowid = products_fts.rowid
        WHERE products_fts MATCH :match
        ORDER BY rank            -- FTS5 rank (bm25); for FTS4 compute from matchinfo
        LIMIT 50
    """)
    suspend fun search(match: String): List<ProductResult>

    @RawQuery suspend fun rebuild(query: SupportSQLiteQuery): Int   // INSERT INTO
products_fts(products_fts) VALUES('rebuild')
}
```

> With **FTS5**, use `bm25(products_fts, 10.0, 5.0, 1.0)` to weight columns (title 10×, brand 5×, description 1×) and `highlight()` / `snippet()` for excerpts. **FTS4** (broadly supported by Room) ranks via `matchinfo()` / `offsets()`; compute a score in Kotlin or a helper.

35.7.3 Query sanitizer (user text → safe MATCH syntax)

```
object FtsQuery {
    // Turn free text into a safe prefix MATCH query; escape FTS specials.
    fun toMatch(input: String): String {
        val terms = input.trim().split(Regex("\\s+"))
            .filter { it.isNotBlank() }
            .map { sanitizeToken(it) }
            .filter { it.isNotEmpty() }
        if (terms.isEmpty()) return ""
        // Last term gets a prefix star for typeahead; quote to avoid operator injection.
        return terms.mapIndexed { i, t ->
            if (i == terms.lastIndex) "\"$t\"*" else "\"$t\""
        }.joinToString(" ")                                    // implicit AND
    }
    private fun sanitizeToken(t: String) = t.replace("\"", "").filter { it.isLetterOrDigit() }
}
// "running sho" -> "\"running\" \"sho\"*"
```

35.7.4 Index sync (external-content tables need explicit sync)

```
// For external-content FTS, writes to the source must be mirrored into the index.
// Easiest: do both in one DAO transaction, or use SQL triggers created in a Room migration.
@Dao
interface ProductWriteDao {
    @Transaction
    suspend fun upsertAndIndex(entity: ProductEntity) {
        upsert(entity)
        // Keep FTS in sync (for content tables, 'rebuild' or targeted insert/delete).
        syncFts(entity.rowidOrSelect())
    }
    @Query("INSERT INTO products_fts(products_fts, rowid, title, brand, description) " +
           "VALUES('delete', :rowid, '', '', '')")   // remove old, then re-add (illustrative)
    suspend fun ftsDelete(rowid: Long)
    // In practice: define AFTER INSERT/UPDATE/DELETE triggers on `products` to maintain
`products_fts`.
}
```

> **Simplest robust approach:** create SQLite **triggers** (`AFTER INSERT/UPDATE/ DELETE` on `products`) that maintain `products_fts` , added in a Room migration (Ch 36). Then normal Room writes keep the index current automatically.

35.7.5 Merge local + remote (Ch 10)

```
// Offline-first search: local FTS instantly, then merge remote when online.
fun search(query: String): Flow<SearchUiState> = flow {
    val match = FtsQuery.toMatch(query)
    val local = searchDao.search(match).map { it.toDomain() }
    emit(SearchUiState.Results(local, loadingRemote = connectivity.isOnline))   // local-first
(Ch 10)
    if (connectivity.isOnline) {
        val remote = remoteSearch(query)                // Ch 9
        emit(SearchUiState.Results(merge(local, remote), loadingRemote = false))
    }
}.flowOn(dispatchers.io)
```

35.7.6 Module structure

```
:data:catalog/search  ProductFts, ProductSearchDao, FtsQuery, merge (Ch 10)
:core:database        FTS triggers (migration, Ch 36)
:feature:search       uses local+remote (Ch 9/10)
```

35.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Index out of sync with source	stale/missing results	triggers maintain FTS; or atomic write-both in a transaction
Search during bulk insert	partial results	index in the same transaction; or rebuild after bulk load
Rapid queries (typeahead)	wasted work	debounce (Ch 9) + `flatMapLatest`
Unsanitized input	FTS syntax error/injection	sanitize to valid MATCH; quote terms
Large rebuild	jank	rebuild off main thread; incremental sync preferred
Concurrent read/write FTS	lock contention	SQLite serializes; keep writes batched

- **Index sync is the core correctness concern:** use triggers (automatic) or transactional write-both so the FTS index never drifts from the source table.
- **Sanitize queries** — raw user input can contain FTS operators (`*` , `"` , `-` , `NEAR`) that cause syntax errors or unexpected matches.

35.9 Failure Scenarios and Recovery

Failure	Detection	Response	Outcome
Query syntax error (specials)	SQLiteException	sanitize input; fall back to safe terms	no crash
Index drift (missing results)	QA	triggers / rebuild index	coherent
Offline	connectivity	serve local FTS only (merge skips remote)	works offline
No local results	empty	show remote (if online) / empty state	graceful
FTS5 unavailable on device	build/runtime	use FTS4 (Room-supported)	compatible
Bulk sync invalidates index	—	rebuild or incremental update post-sync	fresh
Huge index	storage	external-content (no dup); prune; bound	bounded
Tokenizer mismatch (diacritics)	poor matches	`remove_diacritics` / `unicode61` options	better recall

 35.9 Failure Scenarios and Recovery

> **FTS4 vs FTS5 compatibility:** Room has first-class `@Fts4` support across Android versions; FTS5 depends on the device's bundled SQLite. For broad reach, **FTS4** is the safe default; use FTS5 (better ranking/snippet) when you control the SQLite version (e.g., bundling a SQLite library).

35.10 Scaling the Implementation

- **Level 1.** `LIKE` for a tiny table (acceptable < a few hundred rows).
- **Level 2.** Room `@Fts4` external-content index + MATCH queries + sanitizer.
- **Level 3 — production.** Ranking (BM25/matchinfo) + field weights + snippet; triggers for index sync; prefix typeahead; merge local+remote (Ch 10).
- **Level 4 — large.** FTS5 with `bm25()` column weights; `remove_diacritics` /custom tokenizer; incremental index updates from sync (Ch 32); paging results (Ch 13).
- **Level 5 — enterprise.** Hybrid local+server search ranking, synonyms/stemming, multi-language tokenizers, search analytics, on-device embeddings/semantic search.

Avoid overengineering: a small bounded dataset can use `LIKE`. Reach for FTS when the dataset is large, offline search matters, or you need ranking/prefix/snippet.

35.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
Local search	FTS (inverted index)	`LIKE '%q%'`	FTS for large/ranked/offline; `LIKE` only for tiny tables
FTS version	FTS4 (Room-native, broad)	FTS5 (bm25/snippet, needs newer SQLite)	FTS4 for compatibility; FTS5 when you control SQLite
Storage	External-content table	contentless/own copy	External-content avoids duplicating data; own copy if source isn't in SQLite
Index sync	SQL triggers	repo write-both	Triggers (automatic, robust); repo for simple cases
Ranking	BM25 (FTS5)	matchinfo custom (FTS4)	BM25 easiest where available; matchinfo for FTS4
Search scope	Local-only	Local + remote merge (Ch 10)	Merge for offline-first; local-only if remote not needed

FTS vs `LIKE`: `LIKE '%q%'` can't use an index for a leading wildcard (full scan), can't rank, and handles only substrings; FTS builds an inverted index for fast tokenized, rankable, prefix/phrase search. For anything beyond a tiny table or exact-prefix (`LIKE 'q%'` can index), use FTS.

FTS4 vs FTS5: FTS5 has better ranking (`bm25`), `snippet()` / `highlight()`, and configurability, but depends on the device's SQLite version; Room's `@Fts4` works everywhere. Choose FTS4 for maximum compatibility, FTS5 (often via a bundled SQLite) when ranking/snippet quality matters.

35.12 Security Considerations

- **Sanitize query input** to valid MATCH syntax — prevents FTS syntax errors and avoids surprising matches; it's input handling, not SQL injection (Room parameterizes), but still validate.
- **Search is over local cached data** — if that data is sensitive, the FTS **index is also sensitive** and sits in the DB: encrypt the DB (Ch 24) so the index can't be read off a stolen device.
- **Per-account scoping / clear on logout** (Ch 22/42): the index contains the user's data; clear it on logout.
- **Query text is PII-ish** (names, private terms) — redact in logs (Ch 28); don't send local queries to analytics without consent.
- Client vs backend: local FTS searches only what the user is authorized to have cached; the backend owns authoritative search over data the client doesn't hold.

35.13 Performance and Resource Optimization

- **Index, don't scan:** FTS MATCH uses the inverted index — orders of magnitude faster than `LIKE` on large tables.
- **External-content** avoids duplicating the data (storage) — the index references source rows.
- **Debounce** + `flatMapLatest` (Ch 9) to avoid per-keystroke queries; run on IO.
- **Limit + page** results (Ch 13) for large result sets.
- **Incremental index updates** (triggers) over full rebuilds; rebuild only after bulk loads.

- **Tokenizer cost** is negligible; choose options (`remove_diacritics`) for recall.
- **Profiling:** query latency (FTS vs LIKE), index size, index-sync cost, Database Inspector `EXPLAIN QUERY PLAN`.

35.14 Testing Strategy

In-memory Room with the FTS table; assert matching, ranking, prefix, sync.

```
@RunWith(AndroidJUnit4::class)
class FtsSearchTest {
    private lateinit var db: AppDatabase
    @Before fun setUp() {
        db = Room.inMemoryDatabaseBuilder(ctx, AppDatabase::class.java).build()
        db.productWriteDao().apply {
            upsertAndIndex(product("1", title = "Running Shoe", brand = "Acme"))
            upsertAndIndex(product("2", title = "Dress Shirt", brand = "Acme"))
        }
    }
    @After fun tearDown() = db.close()

    @Test fun `token match finds the right row`() = runTest {
        val r = db.searchDao().search(FtsQuery.toMatch("running"))
        assertEquals(listOf("1"), r.map { it.id })
    }
    @Test fun `prefix typeahead matches`() = runTest {
        assertEquals(listOf("1"), db.searchDao().search(FtsQuery.toMatch("runni")).map { it.id
})
    }
    @Test fun `multi-term is AND`() = runTest {
        db.productWriteDao().upsertAndIndex(product("3", title = "Running Shirt"))
        val r = db.searchDao().search(FtsQuery.toMatch("running shirt"))
        assertEquals(listOf("3"), r.map { it.id })        // both terms
    }
    @Test fun `index stays in sync on update`() = runTest {
        db.productWriteDao().upsertAndIndex(product("1", title = "Trail Boot"))   // retitle
        assertTrue(db.searchDao().search(FtsQuery.toMatch("running")).isEmpty())  // old term
gone
        assertEquals(listOf("1"), db.searchDao().search(FtsQuery.toMatch("trail")).map { it.id
})
    }
    @Test fun `sanitizer neutralizes FTS operators`() {
        assertFalse(FtsQuery.toMatch("red\" OR 1").contains("OR 1"))              // no
injection
    }
}
```

Also: ranking/field-weight order (FTS5), snippet output, diacritics/case, offline-only merge (Ch 10), large-table performance benchmark.

35.15 Production Observability

- **Metrics:** local-search latency (vs remote), local-result rate (how often local satisfies), index size, index-sync cost, zero-local-result rate, FTS4-vs-FTS5 usage.
- **Logs:** `local_search{q_len, results, latency_ms}` (no raw query — PII, Ch 28).
- **Debugging:** missing results → index drift (fix triggers/sync); slow search → `LIKE` not FTS, or no limit; syntax crashes → unsanitized input; poor recall → tokenizer (diacritics/case); FTS5 crash on old device → use FTS4.
- **Privacy:** index + queries are user data; encrypt DB (Ch 24); redact logs.

35.16 System Design Interview Preparation

> ≥25 questions.

Fundamentals

1. *Why not* `LIKE '%q%'` *?* Full scan (no index for leading wildcard), no ranking, no tokenization/prefix/phrase.
2. *What is FTS?* SQLite full-text search — an inverted index enabling fast tokenized, rankable search.
3. *Room FTS?* `@Fts4` / `@Fts5` virtual tables; query with `MATCH`.
4. *Why local FTS?* Offline search over cached data (Ch 29).
5. *Inverted index?* term → documents mapping; matches by token.

Design 6. *External-content table?* FTS indexes an existing table without duplicating data (`content=`). 7. *Index sync?* Triggers (AFTER INSERT/UPDATE/DELETE) or transactional write-both. 8. *Tokenizer?* Controls splitting/normalization (case, diacritics) — `unicode61`, `remove_diacritics`. 9. *Query syntax?* Prefix (`shoe*`), phrase (`"a b"`), boolean (AND/NOT), column filters.

Ranking/UX 10. *Ranking?* FTS5 `bm25()` (weight columns: title > description); FTS4 `matchinfo`. 11. *Snippet/highlight?* FTS5 `snippet()` / `highlight()`. 12. *Typeahead?* Prefix `term*`; debounce (Ch 9). 13. *Merge local+remote?* Local-first instant, then remote merge (Ch 10).

FTS4 vs FTS5 14. *Difference?* FTS5 has bm25/snippet/config but needs newer SQLite; FTS4 is Room-native/broadly supported. 15. *Which to pick?* FTS4 for compatibility; FTS5 when you control SQLite/need ranking.

Correctness 16. *Index drift?* Triggers / atomic write keep FTS in sync. 17. *Sanitize queries?* Yes — neutralize FTS operators; valid MATCH. 18. *Bulk load?* Rebuild or incremental sync after.

Security 19. *Sensitive data?* Encrypt the DB (index included, Ch 24). 20. *Per-account?* Clear index on logout (Ch 22/42). 21. *Query PII?* Redact logs (Ch 28).

Performance 22. *Why FTS fast?* Inverted index vs full scan. 23. *Storage?* External-content avoids duplication. 24. *Large results?* Limit + page (Ch 13).

Testing 25. *How test?* In-memory Room; assert token/prefix/multi-term match + sync. 26. *How test sanitizer?* Assert operators neutralized.

Production debugging 27. *Missing results?* Index drift. 28. *Crash on query?* Unsanitized input. 29. *Poor recall?* Tokenizer/diacritics.

Design-change challenges

1. **Search catalog offline.** `@Fts4` external-content + MATCH + sync triggers.
2. **Rank title over description.** FTS5 `bm25()` column weights.
3. **Typeahead over local data.** Prefix `term*` + debounce (Ch 9).
4. **Highlight matches.** FTS5 `snippet()` / `highlight()`.
5. **Merge with remote search.** Local-first + remote (Ch 10).

Mock walkthrough (abridged)

Clarify (dataset size, offline, ranking, prefix?) → FTS over `LIKE` → `@Fts4` / `@Fts5` external-content → MATCH (prefix/phrase) + sanitize → bm25/field weights + snippet → index sync (triggers) → merge local+remote (Ch 10) → security (encrypt/clear) → testing.

35.17 Practical Exercises

1. **Beginner.** Add a Room `@Fts4` external-content table over a products table; search by token. *Eval:* matches words, not substrings.
2. **Intermediate.** Add a query sanitizer with prefix typeahead and multi-term AND; keep the index synced via triggers. *Eval:* prefix works; index stays current.
3. **Advanced.** Rank results (bm25/field weights) and produce highlighted snippets; merge with remote (Ch 10). *Eval:* relevant order; highlights; offline-first.
4. **Production debugging.** Fix slow `LIKE` search and an index that drifts on update. *Eval:* FTS-fast; coherent index.
5. **System design challenge.** Encrypt the DB (index included), clear on logout, and add diacritic-insensitive matching. *Eval:* secure; per-account; better recall.

(Solutions withheld unless requested.)

35.18 Chapter Summary & Reference Sheet

Key decisions: use **SQLite FTS via Room** (`@Fts4` for broad compatibility, `@Fts5` where you control SQLite) over `LIKE` for local search; use an **external- content** table to index the source without duplicating data; query with **MATCH** (prefix/phrase/boolean), **rank with BM25 + field weights**, and produce **snippet/ highlight**; **keep the index in sync** via SQLite **triggers** (or transactional write-both); **sanitize** user input to valid MATCH; **merge local-first with remote** (Ch 10) for offline-first search; **encrypt** the DB (index included, Ch 24) and **clear per-account** (Ch 22/42).

Concepts: inverted index, FTS4 vs FTS5, external-content tables, tokenizers, MATCH syntax (prefix/phrase/boolean), BM25 ranking + field weights, snippet/ highlight, index sync (triggers), local+remote merge.

Common mistakes: `LIKE '%q%'` on large tables (slow, no rank); duplicating data (no external-content); index drift (no triggers/sync); unsanitized queries (syntax errors); FTS5 on devices without it (use FTS4); unencrypted sensitive index; not clearing on logout.

Checklist: ☐ FTS (not LIKE) ☐ FTS4/5 chosen for compatibility ☐ external-content table ☐ MATCH (prefix/phrase) ☐ sanitize input ☐ BM25 + field weights (FTS5) ☐ snippet/highlight ☐ index sync via triggers ☐ merge local+remote (Ch 10) ☐ encrypt DB + clear per-account.

Key APIs: Room `@Fts4(contentEntity=...)` / `@Fts5`, `MATCH`, `bm25()` / `snippet()` / `highlight()` (FTS5), `matchinfo()` / `offsets()` (FTS4), `unicode61` / `remove_diacritics` tokenizer, SQLite triggers (migration, Ch 36), `@RawQuery` rebuild.

References:

- Room FTS: https://developer.android.com/training/data-storage/room/defining-data#fts
- SQLite FTS5: https://www.sqlite.org/fts5.html
- SQLite FTS4: https://www.sqlite.org/fts3.html
- (composes) Ch 9 Debounced Search, Ch 10 Suggestions, Ch 29 Offline-First, Ch 24 Encryption

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/failure diagrams; FTS entity + search DAO + sanitizer + index sync + merge; concurrency (index sync, sanitize); failure (syntax/drift/FTS5 compat); 5 levels; FTS-vs-LIKE and FTS4-vs-FTS5 trade-offs; security (encrypt index/clear); perf (index vs scan); in-memory Room tests; observability (local-result rate); 29 Q&A + 5 challenges. ✅
- **Expand later:** semantic/on-device search, synonyms/stemming, multi-language tokenizers, migrations for triggers (Ch 36).
- **Interview coverage:** exceeds ≥25 bar.